

3. The halogens are a group of elements found in group 7 of the periodic table. The table below gives some information about the elements in this group.

Halogen	Atomic number	Melting point (°C)	Boiling point (°C)	Radius of the atom (units)
fluorine	9	-219	-188	(0.64) / 1.1 / 1.7
chlorine	17	(-5 / -101 / -220)	-34	0.99
bromine	35	-7	59	(1.14 / 1.45 / 1.56)
iodine	(22 / 31 / 53)	114	184	1.33
astatine	85	302	380	1.40

- (a) **Complete** the table above by **circling** the correct value in each of the brackets. One has been done as an example. [3]

- (b) Use the information in the table and your own knowledge to answer the following questions.

- (i) The boiling points of the alkali metals (group 1) decreases from the top to the bottom of their group. Describe whether the halogens follow the same trend. [2]

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- (ii) State the **two** halogens in the table which are gases at 20°C. [2]

..... and

- (iii) State which halogen is a liquid at 20°C. [1]

.....

- (iv) State what happens to the size of the halogen atoms as you go down from the top to the bottom of the group. [1]

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(v) **Tick (✓)** the box next to the correct statements about halogens. [3]

Fluorine has the highest melting point

☐

The most reactive halogen is fluorine

☐

Astatine has the most electrons

☐

Iodine is a solid at 20°C

☐

The most reactive halogen is astatine

☐

Chlorine boils at a lower temperature than fluorine

☐

(c) Sodium chloride can be made by burning sodium in chlorine gas. However it is found naturally in rock salt, which can be obtained from the Earth's crust. **Tick (✓)** the box next to the correct process for extraction of rock salt from the Earth's crust. [1]

fracking

☐

deep shaft mining

☐

fractional distillation

☐

drilling

☐